Available online at www.sciencedirect.com
ScienceDirect
Journal of Electromyography and Kinesiology 18 (2008) 54–67
JOURNAL OF ELECTROMYOGRAPHY AND KINESIOLOGY
www.elsevier.com/locate/jelekin

Time and frequency domain responses of the mechanomyogram and electromyogram during isometric ramp contractions: A comparison of the short-time Fourier and continuous wavelet transforms

Eric D. Ryan *, Joel T. Cramer, Alison D. Egan, Michael J. Hartman, Trent J. Herda

Department of Health and Exercise Science, University of Oklahoma, 1401 Asp Avenue, Norman, OK 73019-6081, USA

Received 24 July 2006; received in revised form 5 September 2006; accepted 6 September 2006

Abstract

The purposes of this study were to examine the mechanomyographic (MMG) and electromyographic (EMG) time and frequency domain responses of the vastus lateralis (VL) and rectus femoris (RF) muscles during isometric ramp contractions and compare the time–frequency of the MMG and EMG signals generated by the short-time Fourier transform (STFT) and continuous wavelet transform (CWT). Nineteen healthy subjects (mean ± SD age = 24 ± 4 years) performed two isometric maximal voluntary contractions (MVCs) before and after completing 2–3, 6-s isometric ramp contractions from 5% to 100% MVC with the right leg extensors. MMG and surface EMG signals were recorded from the VL and RF muscles. Time domains were represented as root mean squared amplitude values, and time–frequency representations were generated using the STFT and CWT. Polynomial regression analyses indicated cubic increases in MMG amplitude, MMG frequency, and EMG frequency, whereas EMG amplitude increased quadratically. From 5% to 24–28% MVC, MMG amplitude remained stable while MMG frequency increased. From 24–28% to 76–78% MVC, MMG amplitude increased rapidly while MMG frequency plateaued. From 76–78% to 100% MVC, MMG amplitude plateaued (VL) or decreased (RF) while MMG frequency increased. EMG amplitude increased while EMG frequency changed only marginally across the force spectrum with no clear deflection points. Overall, these findings suggested that MMG may offer more unique information regarding the interactions between motor unit recruitment and firing rate that control muscle force production during ramp contractions than traditional surface EMG. In addition, although the STFT frequency patterns were more pronounced than the CWT, both algorithms produced similar time–frequency representations for tracking changes in MMG or EMG frequency.
© 2006 Elsevier Ltd. All rights reserved.

Keywords: MMG; EMG; Motor unit recruitment; Motor unit firing rate; Rate coding

1. Introduction

The mechanomyogram (MMG) is commonly defined as the recording of the low-frequency lateral oscillations of active skeletal muscle fibers (Orizio, 1993) and is considered to be the "mechanical counterpart" of motor unit electrical activity measured by the surface electromyogram (EMG). Barry and Cole (1990) and Orizio (1993) have suggested that the lateral oscillations recorded by MMG are a function of: (a) gross lateral movements of the muscle at the onset of contraction due to non-simultaneous activation of muscle–fibers, (b) smaller subsequent lateral oscillations occurring at the resonant frequency of the muscle, and (c) dimensional changes of the active muscle fibers (Barry and Cole, 1990; Orizio, 1993; Orizio et al., 2003).

Like surface EMG, the MMG signal represents a summation of the mechanical activity from individual motor units (Orizio et al., 2003). During voluntary contractions, it has been suggested that the time and frequency domain parameters of the MMG reflect motor unit recruitment and firing rate, respectively (Beck et al., 2006b; Orizio, 1993; Orizio et al., 2003). This is in contrast with the time and frequency domain parameters of the surface EMG

* Corresponding author. Tel.: +1 405 325 5211; fax: +1 405 325 0594.
E-mail address: eryan@ou.edu (E.D. Ryan).

1050-6411/$ - see front matter © 2006 Elsevier Ltd. All rights reserved.
doi:10.1016/j.jelekin.2006.09.003

signal, which are thought to reflect muscle activation (a combination of motor unit recruitment and firing rate) and motor unit action potential conduction velocity, respectively (Stulen and DeLuca, 1981). Therefore, due to their unique and complimentary capabilities, simultaneous measurements of MMG and EMG have been used to identify the dissociation between the electrical and mechanical events (excitation–contraction coupling) that occur with fatigue (Stokes and Dalton, 1991) and to examine factors related to electromechanical and phonomechanical delay (Petitjean et al., 1992).

Several previous studies have also utilized MMG and EMG to characterize the motor control strategies that modulate muscle force production. Most of these investigations have performed isometric "step" contractions (Beck et al., 2005c, 2004; Bilodeau et al., 1991; Coburn et al., 2005, 2004; Dalton and Stokes, 1993; Ebersole et al., 1998, 1999; Lariviere et al., 2001; Maton et al., 1990), which involve separate isometric contractions performed at specific force increments from submaximal to maximal levels of force production (i.e., 20%, 40%, 60%, 80%, 100% MVC, etc.). The same step contraction principle has also been applied to dynamic muscle actions (Beck et al., 2004; Beck et al., 2004; Coburn et al., 2005; Coburn et al., 2004), which may employ different motor control strategies (Kossev and Christova, 1998). Fewer studies have utilized isometric "ramp" contractions (Akasaka et al., 1997, 2004, 2002, 2001, 2003; Bilodeau et al., 1991; Lariviere et al., 2001), which consist of a single, non-fatiguing, linearly increasing, force-varying contraction over a 5–7 second period (Akasaka et al., 1997; Akataki et al., 2004, 2002, 2001, 2003; Bilodeau et al., 1991; Lariviere et al., 2001). Akataki and coworkers Akataki et al., 2004; Akataki et al., 2001; Akataki et al., 2003) and Orizio et al. (2003) have suggested that isometric ramp contractions may provide higher resolution throughout the force spectrum, shorter time taken for data acquisition, and less susceptibility to fatigue than step contractions. It is possible, however, that ramp and step contractions that characterize the force spectrum may incorporate different motor control strategies, much like the comparisons between isometric and dynamic isokinetic step contractions (Beck et al., 2004,; Coburn et al., 2005, 2004). To our knowledge, only two studies have examined the simultaneous measurement of MMG and EMG during isometric ramp contractions in the biceps brachii (BB) (Akataki et al., 2004; Orizio et al., 2003), and no studies have examined these relationships for the superficial quadriceps femoris muscles during leg extension muscle actions.

Bilodeau et al. (1991, 1997) have examined the issues surrounding the use of traditional Fourier-based procedures for determining the center frequency of discrete EMG signal portions extracted from a non-stationary isometric ramp contraction. It has been suggested that the EMG signal epoch being analyzed with a Fourier transform must remain stationary (or at least quasi-stationary) in the time and frequency domains in order to extract meaningful frequency information from the power spectrum (Bonato et al., 1996). Bonato et al. (1996) have proposed that a surface EMG signal epoch can be considered quasi-stationary if it is recorded during a constant–force contraction and is 0.5–2.0 s in duration. Unfortunately, at least one, if not both, conditions are not met during an isometric ramp contraction. Despite the force-varying non-stationarity of ramp contractions, Bilodeau et al. (1997) suggested that local stationarity exists for portions of the ramp contraction selected over small force increments during slow force increment rates. However, to our knowledge, no previous studies have compared the EMG or MMG frequency patterns of response during ramp contractions computed by traditional Fourier procedures, such as the short-time Fourier transform (STFT), and more computer-intensive time–frequency analyses, such as the continuous wavelet transform (CWT), which was proposed to be superior to the STFT (Karlsson et al., 1999, 2000). Therefore, the purposes of the present study were two-fold: (a) to extend the findings of Akataki and colleagues (Akataki et al., 2004; Akataki et al., 2002; Akataki et al., 2001; Akataki et al., 2003) and Orizio et al. (2003) and examine the MMG and EMG time and frequency domain responses of the vastus lateralis (VL) and rectus femoris (RF) muscles during isometric ramp contractions of the leg extensors and (b) to compare the time–frequency representations of the MMG and EMG signals generated by the STFT and CWT procedures during isometric ramp contractions.

2. Methods

2.1. Participants

Nineteen healthy subjects (mean ± SD age = 24 ± 4 years; height = 173 ± 9 cm; mass = 75 ± 13 kg) volunteered for this investigation. None of the participants reported any current or ongoing neuromuscular diseases or musculoskeletal problems specific to the ankle, knee, or hip joints. Each participant completed a pre-exercise health and exercise status questionnaire and signed a written informed consent document. Fifteen of 19 participants reported engaging in 1.5–7 h of aerobic exercise, 12 of 19 reported 1.5–7 h of resistance exercise, and 7 of 19 reported 1–4 h of recreational sports per week, however, none of the participants were competitive athletes. This study was approved by the University Institutional Review Board for Human Subjects Research.

2.2. Isometric assessments

Isometric torque for the right leg extensor muscles was measured using a Biodex System 3 isokinetic dynamometer (Biodex Medical Systems, Inc., Shirley, NY). The participants were seated with restraining straps over the pelvis, trunk, and contralateral thigh, and the lateral condyle of the femur was aligned with the input axis of the dynamometer in accordance with the Biodex User's Guide

(Biodex Pro Manual, Applications/Operations. Biodex Medical Systems, Inc., Shirley, NY. 1998). All isometric torque assessments were performed at a leg flexion angle of 60° below the horizontal plane.

Each participant visited the laboratory twice, once for familiarization and once for the experimental trail. The familiarization trial included two 3-s isometric maximally voluntary contractions (MVCs), at least five isometric ramp contractions, and two additional 3-s isometric MVCs. The isometric ramp contractions were performed until the participant successfully completed the protocol (based on visual inspection) within a maximum of 12–15 attempts. The purpose of the familiarization trial was for the participants to gain experience with the testing equipment and to practice the MVC and ramp contractions. Within 2–5 days after the familiarization trial, participants reported to the laboratory for the experimental trial, which consisted of the following procedures (in order): (a) 2 isometric MVCs, (b) 2–3 isometric ramp contractions and (c) 2 isometric MVCs.

Following a 5-min warm-up on a stationary cycle ergometer (Monark Ergomedic 828 E, Vansbro, Sweden) with the workload set at 50 W and a pedaling cadence of 50–60 rpm, participants performed two 3-s isometric MVCs before the first and after the last isometric ramp contraction. Participants were asked to produce as much force as possible for 3-s and strong verbal encouragement was provided. Two minutes of rest was allowed between each MVC trial. Peak MVC torque was calculated as the highest average torque value that occurred during any 0.5-s duration within the 3-s MVC. The highest MVC torque value recorded for the 2 MVC trials was used as the representative value and as the 100% torque value during the subsequent isometric ramp contractions.

For the isometric ramp contractions, participants were required to track their torque production on a computer monitor placed in front of them that displayed the real-time, digitized torque signal overlayed onto a programmed ramp template. The ramp template consisted of a 5-s horizontal baseline at 5% MVC followed by a linearly increasing ramp line lasting 6 s and ending with a 2-s horizontal plateau at 100% MVC (Bilodeau et al., 2003). During the experimental trial, participants were asked to perform 2–3 ramp contractions (5–100% MVC) and verbal encouragement was provided. A 2-min rest period was allowed between each ramp contraction. The ramp trials that best satisfied the following criteria were used for later analysis: (a) maximal force reaching at least 85% of MVC and (b) a maximum tracking error of ± 3% MVC around the template. The isometric ramp template and real-time torque overlay was programmed using LabVIEW 7.1 software (National Instruments, Austin, TX).

2.3. Mechanomyography

The MMG signals were detected with active miniature accelerometers (EGAS-FS-10-/VO5, Measurement Specialties, Inc., Hampton, VA) that were pre-amplified with a gain of 200, frequency response of 0–200 Hz, sensitivity of 70 mV/m s^{-2}, and range of ±98 m s^{-2}. The placement for the vastus lateralis (VL) was over the lateral/anterior portion of the muscle at 50% the distance between the greater trochanter and lateral condyle of the femur. The placement for the rectus femoris (RF) was at the midpoint between the inguinal crease and the superior border of the patella. The accelerometers were fixed to the skin with 3M double-sided foam tape (Fig. 1).

2.4. Surface electromyography

Bipolar surface electrodes were placed along the longitudinal axes of the VL and RF muscles (Fig. 1). Local areas of the skin where the MMG and EMG sensors were placed were shaved and cleaned with isopropyl alcohol. The EMG electrode arrangement was placed so that the midpoint of the center-to-center interelectrode distance was 25 mm distal to the MMG sensor. The EMG signals were recorded with pre-amplified, active electrodes (TSD150B, Biopac Systems Inc., Santa Barbara, CA) with a fixed center-to-center interelectrode distance of 20 mm, built-in differential amplifier with a gain of 350 (nominal), input impedance of 100 MΩ, common mode rejection ratio of 95 dB (normal), bandwidth of 12–500 Hz, electrode diameter of 11.4 mm, sensor dimensions of 17.4 mm (wide) × 51 mm (long) × 6.4 mm (high), and sensor mass of 9.5 g. The MMG and EMG sensors were interfaced with a high level transducer module (HLT100C, Biopac Systems, Inc., Santa Barbara, CA). A single pre-gelled, disposable electrode (Ag-Ag Cl, Quinton Quick Prep, Quinton Instruments Co., Bothell, WA) was placed on the cleaned and lightly abraded skin over the spinous process of the 7th cervical vertibrae to serve as a reference electrode.

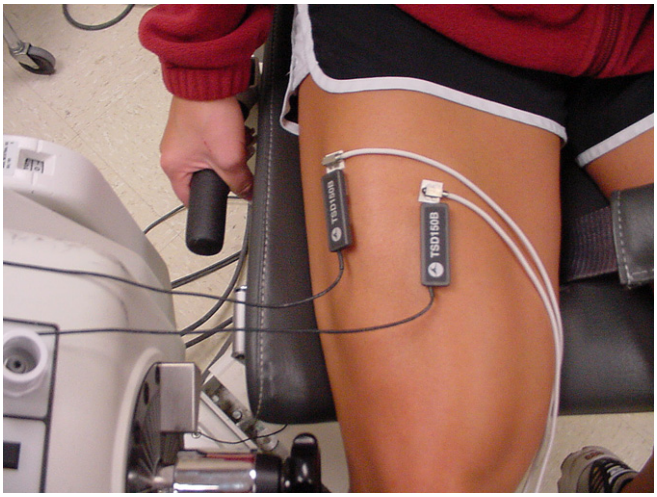


Fig. 1. An example of the mechanomyographic (MMG) sensor and electromyographic (EMG) electrode placements.

2.5. Signal processing

The MMG, EMG, and torque signals were recorded simultaneously with a Biopac data acquisition system (MP150WSW, Biopac Systems, Inc., Santa Barbara, CA), during each isometric MVC and ramp contraction. The torque (Nm) signal from the Biodex dynamometer, and the EMG (μV) and MMG (m s^{-2}) signals recorded from the VL and RF were sampled at 1 KHz. All signals were stored on a personal computer (Dell Inspiron 8200, Dell, Inc., Round Rock, TX), and all processing was completed off-line using custom written software (LabVIEW v 7.1, National Instruments, Austin, TX). The MMG signals were digitally filtered (zero-phase 4th-order Butterworth filter) with a pass band of 5–100 Hz, while the torque signal was low-pass filtered with a 10 Hz cutoff (zero-phase 4th-order Butterworth filter). All subsequent analyses in the time and frequency domains were performed on the filtered signals.

MVC torque (Nm) was determined as the highest 0.5-s average torque value that occurred during the middle 2-s of the 3-s MVC. 0.5-s epochs of the MMG and EMG signals for the VL and RF that corresponded with the same 0.5-s epoch used to calculate MVC torque were used to calculate MMG and EMG amplitude and frequency estimates during the MVC. For the isometric ramp contractions, 0.6-s epochs were extracted every 0.1-s for the torque, MMG, and EMG signals (Akataki et al., 2001). Torque was characterized as the mean of each 0.6-s epoch, whereas the time domains for the MMG and EMG signal epochs were represented as root mean square (RMS) amplitude values.

Time–frequency representations for the MMG and EMG signals of the VL and RF muscles were constructed using the short-time Fourier transform (STFT) and the continuous wavelet transform (CWT) algorithms. For the signal, $f(t)$, the STFT can be represented with the following equation (Cohen, 1995):

$$F_{\mathrm{STFT}}(t,\omega) = \int f(t+\tau)h(\tau)\mathrm{e}^{-\mathrm{j}\omega\tau}\,\mathrm{d}\tau$$

where the sliding window function, $h(\tau)$, consisted of 0.6-s epochs taken every 0.1 s and treated with a Hamming window. The power spectral density function for each windowed epoch was estimated with the discrete Fourier transform, rather than the fast Fourier transform, to avoid the constraint of 2^n number of data points. Mean power frequency (MPF) estimates were calculated from the power spectrum of each epoch with the following equation (Basmajian and DeLuca, 1985):

$$f_{\mathrm{mean}} = \frac{\int_0^f f S_{\mathrm{m}}(f)\,\mathrm{d}f}{\int_0^f S_m(f)\,\mathrm{d}f}$$

where $S_m(f)$ is the power spectrum.

The CWT function was computed using the LabVIEW Advanced Signal Processing Toolkit (National Instruments, Austin, TX), which was defined as

$$\mathrm{CWT}(a,\tau) = \frac{1}{\sqrt{a}}\int_{-\omega}^{+\omega} s(t)\Psi_0^*\left(\frac{t-\tau}{a}\right)\mathrm{d}t$$

where $s(t)$ is the input signal, a represents the wavelet scale, τ represents the wavelet translation, and $\psi_0(t)$ is the mother wavelet function. The CWT results in a scalogram that is a two dimensional ($i \times j$) matrix of squared wavelet coefficients consisting of i columns (where i is the number of data points constituting the MMG and/or EMG signals during the ramp contraction) and j rows (where j is the number of wavelet scales). Since the relationship between the wavelet scale and its center frequency is defined as

$$f_a = \frac{f_c}{a}$$

where f_a is the frequency (Hz) corresponding with the wavelet scale a, and f_c is the center frequency of the wavelet at scale $a = 1$, a *Morlet* wavelet ($f_c = 812.5$ Hz) was nonlinearly scaled in the present study so that f_a ranged from 10 to 500 Hz for the EMG signals and 5–100 Hz for the MMG signals in 1 Hz increments. Thus, $j = 490$ for the EMG signals and $j = 95$ for the MMG signals, and the squared wavelet coefficients in each column of the matrix (j), when divided by the corresponding frequency (f_c), represented a power spectrum for each instant in time. The instantaneous mean power frequency (IMPF) values of the time-dependent power spectral density for each MMG and EMG signal, therefore, were calculated according to the following equation (Karlsson and Gerdle, 2001):

$$\mathrm{IMPF}(t) = \frac{\int_0^F \omega P(t,\omega)\,\mathrm{d}\omega}{\int_0^F P(t,\omega)\,\mathrm{d}\omega}$$

where $P(t,\omega)$ is the time-dependent power spectral density, ω is the frequency, F is one-half of the sampling frequency (i.e., Nyquist frequency), and IMPF(t) is the IMPF value at time t. Once the IMPF values were obtained from the CWT across the ramp contraction, 0.6-s sections were averaged every 0.1-s to obtain average IMPF ($\mathrm{CWT_{aIMPF}}$) values that corresponded with the MPF values estimated with the STFT ($\mathrm{STFT_{MPF}}$). The RMS, $\mathrm{STFT_{MPF}}$, and $\mathrm{CWT_{aIMPF}}$ values were then aligned with the corresponding torque values and normalized to the pre-ramp isometric MVC values to create amplitude vs. torque and frequency vs. torque relationships (Akataki et al., 2001). The vertical deflection points in Fig. 3a–f (lines 1, 2, 3, and 4) were determined by the investigators based on the MMG amplitude, MMG $\mathrm{STFT_{MPF}}$, and MMG $\mathrm{CWT_{aIMPF}}$ patterns of response, rather than the curve that best fit the data.

2.6. Statistical analyses

The relationships between normalized amplitude (RMS) vs. normalized torque (% MVC) and normalized frequency ($\mathrm{STFT_{MPF}}$ and $\mathrm{CWT_{aIMPF}}$) vs. % MVC for the MMG and EMG signals from the VL and RF muscles during the isometric ramp contractions were examined using a curve fit

Table 1
Curve fit estimation results for the amplitude (RMS) and frequency ($STFT_{MPF}$ and CWT_{aIMPF}) vs. normalized torque relationships

Dependent variables	3 Highest ranked models					Model selected for interpretation
	Model 1 (R^2)	Model 1 – Model 2 R^2-change	Model 2 (R^2)	Model 2 – Model 3 R^2-change	Model 3 (R^2)	
MMG VL RMS	LIN (0.903)	0.000	QUA (0.903)	0.087	CUB (0.990)[a]	$Y = 60.622 - 2.838X + 0.083X^2 - 0.001X^3$
MMG VL $STFT_{MPF}$	LOG (0.798)	0.009	POW (0.807)	0.046	CUB (0.853)[a]	$Y = 75.786 + 1.492X - 0.025X^2 + 0.0001X^3$
MMG VL CWT_{aIMPF}	LOG (0.885)	0.000	POW (0.885)	0.051	CUB (0.936)[a]	$Y = 78.403 + 1.103X - 0.018X^2 + 0.0001X^3$
MMG RF RMS	EXP (0.867)	0.005	QUA (0.872)	0.121	CUB (0.993)[a]	$Y = 72.987 - 3.615X + 0.119X^2 - 0.001X^3$
MMG RF $STFT_{MPF}$	POW (0.803)	0.000	S (0.803)	0.142	CUB (0.945)[a]	$Y = 47.703 + 2.828X - 0.054X^2 + 0.0003X^3$
MMG RF CWT_{aIMPF}	LOG (0.887)	0.000	POW (0.887)	0.068	CUB (0.955)[a]	$Y = 71.569 + 1.581X - 0.027X^2 + 0.0001X^3$
EMG VL RMS	EXP (0.978)	0.021	QUA (0.999)[a]	0.000	CUB (0.999)	$Y = 8.012 + 0.342X + 0.006X^2$
EMG VL $STFT_{MPF}$	QUA (0.841)	0.013	S (0.854)	0.031	CUB (0.885)[a]	$Y = 77.881 + 0.825X - 0.012X^2 + 0.00005X^3$
EMG VL CWT_{aIMPF}	S (0.827)	0.088	QUA (0.915)	0.035	CUB (0.950)[a]	$Y = 89.310 + 0.431X - 0.006X^2 + 0.00002X^3$
EMG RF RMS	LIN (0.973)	0.025	QUA (0.998)[a]	0.001	CUB (0.999)	$Y = 3.433 + 0.333X + 0.007X^2$
EMG RF $STFT_{MPF}$	POW (0.239)	0.260	QUA (0.499)[a]	0.016	CUB (0.515)	$Y = 92.424 + 0.258\ X - 0.002X^2$
EMG RF CWT_{aIMPF}	S (0.233)	0.428	QUA (0.661)[a]	0.006	CUB (0.667)	$Y = 97.947 + 0.203\ X - 0.002X^2$

All models were statistically significant $P < 0.001$.
CUB – cubic; EXP – Exponential; UN – Linear; LOG – Logarithmic; PCW – Power; QUA-quadratic; S – S-curve.
MMG – Mechanomyography; EMG – Electromyography. VL – Vastus lateralis; RF – Rectus femoris; RMS – Root mean square amplitude;
$STFT_{MPF}$ – shorttime Fourier transform mean power frequency; CWT_{aIMPF} – Confinous wavelet transform average instantaneous mean power frequency.
[a] Indicates the model selected for interpretation for the highest R^2 value that was 0.02 larger than the previously-ranked model.

estimation procedure (SPSS version 12.0, Chicago, IL). All available models were selected during the curve fit estimation, which included linear, quadratic, cubic, exponential, power, logarithmic, and S-curve models. The model selected for interpretation was chosen from the models yielding the three highest R^2 values (Table 1). The models were ranked according to their R^2, and R^2-change was calculated between consecutively-ranked models. A model was selected for interpretation when the highest R^2 value was obtained if the R^2-change was greater than 0.02 from the previously-ranked model. In addition, pre- and post-ramp torque measurements during the MVC trials were analyzed with a paired-samples *t*-test to assess the presence of fatigue. The alpha level was set at $P \leqslant 0.05$ to determine statistical significance.

3. Results

3.1. Torque

The relationship between torque (%MVC) vs. time during the ramp contractions was fit with a linear model ($R^2 = 0.999$; Torque $= 0.481 + 0.020 \times$ time). The tracking error for each individual ramp contraction was kept below 3% MVC. An example of the MMG, EMG, and torque signals during one representative ramp contraction is shown in Fig. 2. In addition, there was no significant difference between the pre- and post-ramp MVC torque values ($P = 0.054$).

3.2. Mechanomyography

The results of the curve fit analysis for the MMG and EMG patterns are presented in Table 1. Fig. 3a and d show the relationships between MMG amplitude (RMS) and torque (% MVC) for the VL and RF muscles that were both fit with cubic models ($R^2 = 0.990$; $R^2 = 0.993$, respectively). The MMG amplitude for the VL muscle did not change from 5% to 28% MVC, increased rapidly from 28% to 78% MVC, and then plateaued from 78% to 100% MVC. MMG amplitude for the RF muscle responded similarly with no change 5% to 24% MVC, increased rapidly from 24% to 76% MVC, plateaued and then decreased from 76% to 100% MVC.

Fig. 3b, c, e, and f show the relationships between MMG frequency ($STFT_{MPF}$ and CWT_{aIMPF}) and torque for the VL and RF muscles that were all fit with cubic models (Table 1). The R^2 values for the $STFT_{MPF}$ and CWT_{aIMPF} patterns of the VL were 0.853 and 0.936, respectively, whereas for the RF the R^2 values were 0.945 and 0.955, respectively. The $STFT_{MPF}$ and CWT_{aIMPF} patterns of response were similar and increased from 5% to 24–28% MVC, plateaued from 24–28 to 76–78% MVC, and then increased again from 76–78 to 100% MVC for both the VL (3B and 3C) and RF (3E and 3F) muscles.

3.3. Electromyography

Fig. 4a and d shows the relationship between EMG amplitude (RMS) and torque (%MVC) for the VL and RF muscles that were both fit with quadratic models ($R^2 = 0.990$; $R^2 = 0.998$, respectively). EMG amplitude for both muscles increased curvilinearly from 5% to 100% MVC for both the VL and RF muscles.

Fig. 4b, c, e, and f shows the relationship between EMG frequency ($STFT_{MPF}$ and CWT_{aIMPF}) and torque for the VL and RF muscles, were the VL was fit with cubic models ($R^2 = 0.885$; $R^2 = 0.950$, respectively), and the RF was fit with quadratic models ($R^2 = 0.499$; $R^2 = 0.661$, respectively). The $STFT_{MPF}$ and CWT_{aIMPF} estimates of EMG

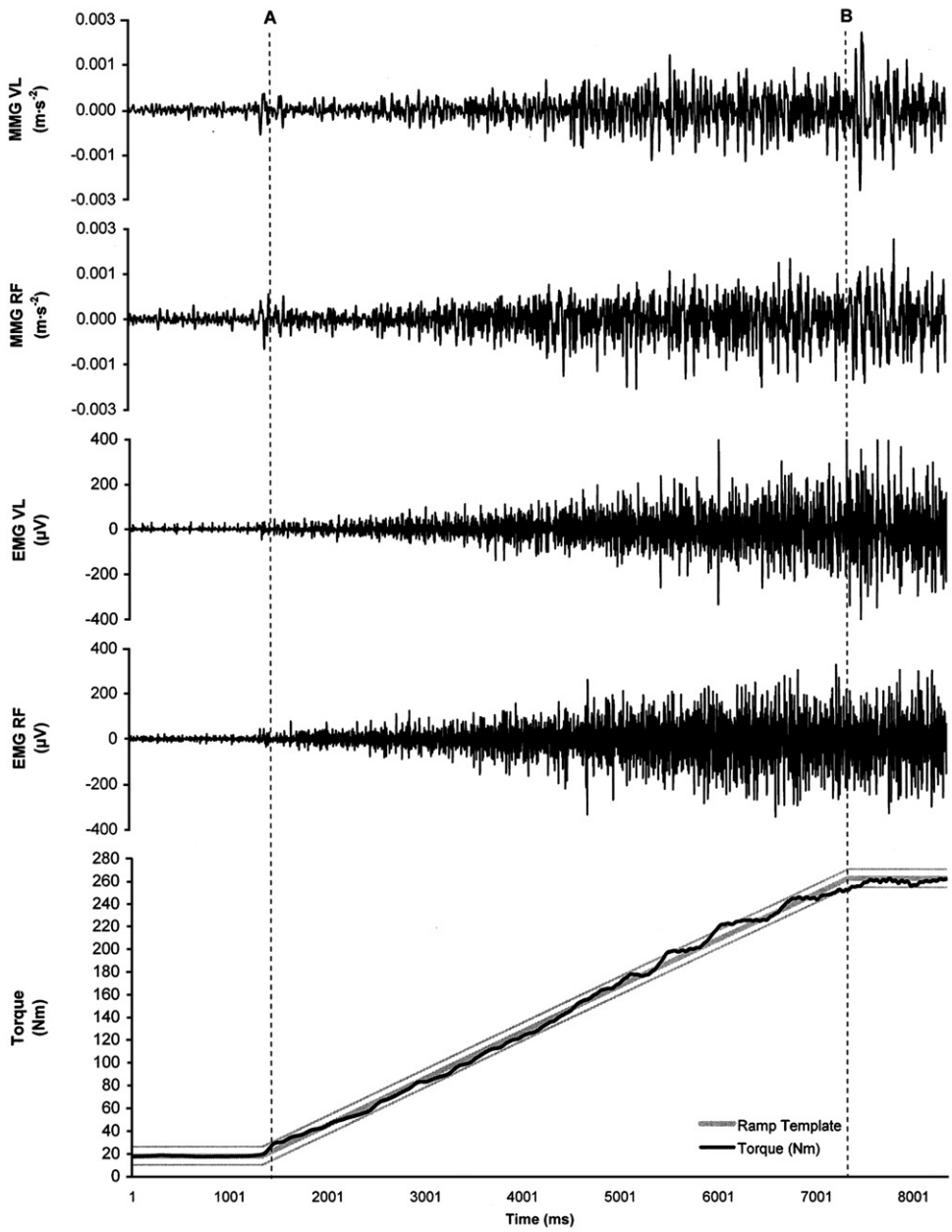


Fig. 2. An example of the mechanomyographic (MMG) and electromyographic (EMG) signals from the vastus lateralis (VL) and rectus femoris (RF) muscles during an isometric ramp contraction for 1 participant. The torque signal (bottom) is overlayed onto the ramp template as it appeared for the participant during the trial. The vertical dashed lines represent the onset (a) and end (b) of the 6-s ramp phase. The MMG and EMG signals that correspond with the 6-s ramp phase (a–b) were selected for analysis.

frequency were similar, but there were no clear deflection points.

4. Discussion

4.1. MMG amplitude

Previous studies have examined the MMG amplitude vs. isometric torque relationship during ramp contractions in the BB muscle up to 70% MVC (Akataki et al., 2003), 80% MVC (Akataki et al., 2004, 2002, 2001), 90% MVC (Orizio et al., 2003), and in the first dorsal interosseous (FDI) up to 70% MVC (Akataki et al., 2003). For the BB, the findings have consistently shown no changes in MMG amplitude from baseline to 20–25% MVC, rapid increases to 60–65% MVC, and decreases from 60% to 65% to the end of the force spectrum (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003). However, for the FDI, MMG amplitude decreased from baseline to 20% MVC, increased from 20% to 40% MVC, and decreased from 40% to 70% MVC (Akataki et al., 2003). Thus, these findings indicated differences in the patterns

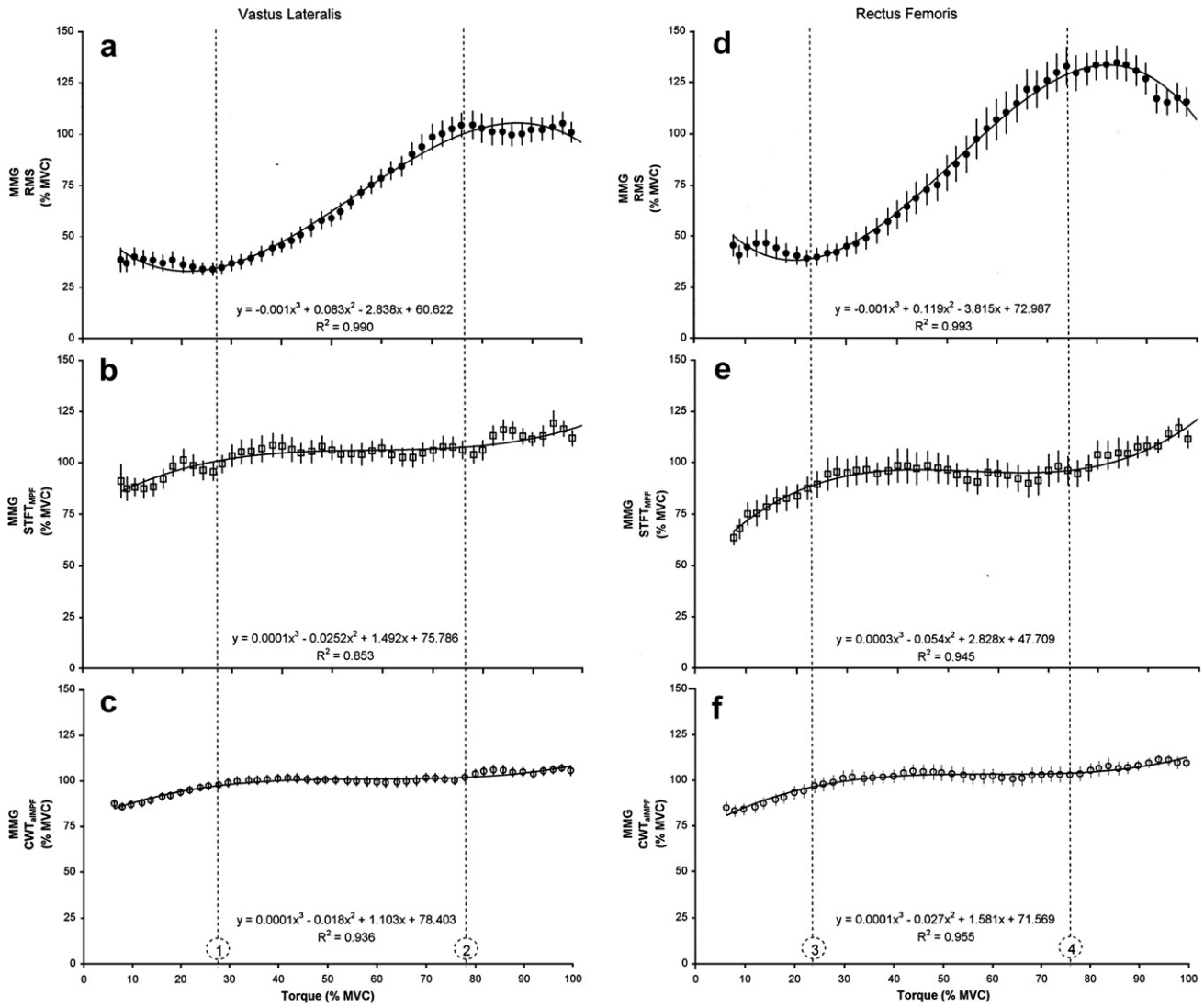


Fig. 3. (a) The relationship between MMG amplitude (RMS, % MVC) and torque (% MVC) for the vastus lateralis (VL) was fit with a cubic model. (b) The relationship between MMG $STFT_{MPF}$ (% MVC) and torque (% MVC) for the VL was fit with a cubic model. (c) The relationship between MMG CWT_{aIMPF} (% MVC) and torque (% MVC) for the VL was fit with a cubic model. (d) The relationship between MMG amplitude (% MVC) and torque (% MVC) for the rectus femoris (RF) was fit with a cubic model. (e) The relationship between MMG $STFT_{MPF}$ (% MVC) and torque (% MVC) for the RF was fit with a cubic model. F. The relationship between CWT_{aIMPF} (% MVC) and torque (% MVC) for the RF was fit with a cubic model. The vertical dashed lines represent deflection points chosen by the investigators based on changes in the MMG amplitude, $STFT_{MPF}$, and CWT_{aIMPF} patterns of response, rather than the curve of best fit. The deflection points occurred at 28% and 78% MVC for the VL and at 24% and 76% MVC for the RF. Error bars reflect standard errors (SE).

of response for MMG amplitude recorded from the BB vs. FDI muscles, which may be explained by the muscle-specific differences in neural control strategies involving motor unit recruitment and firing rate. However, no previous studies have examined the MMG amplitude vs. isometric torque relationship during ramp contractions for any muscles other than the BB and FDI.

Other studies have examined the MMG amplitude vs. isometric torque relationship during incremental step contractions in the BB (Beck et al., 2005c, 2004; Esposito et al., 1996; Maton et al., 1990; Zwarts and Keidel, 1991), superficial quadriceps femoris muscles (Coburn et al., 2005; Coburn et al., 2004; Ebersole et al., 1998; Ebersole et al., 1999; Matheson et al., 1997; Shinohara et al., 1998; Stokes and Dalton, 1991,), and muscles of the hand (Stokes and Cooper, 1992). These studies have reported that the MMG amplitude patterns of response: (a) increases linearly to MVC (Coburn et al., 2004; Orizio et al., 1989; Shinohara et al., 1998; Stokes and Dalton, 1991,; Zwarts and Keidel, 1991), (b) increases curvilinearly to MVC (Maton et al., 1990; Stokes and Cooper, 1992), (c) increases in an S-shaped pattern to MVC (Lammert et al., 1976), or (d) increases to 50–80% MVC (Beck et al., 2005c, 2004; Coburn et al., 2005, 2004,; Ebersole et al., 1998, 1999; Matheson et al., 1997; Orizio et al., 1989) and then plateaus or decreases at higher torque levels. The minor variations

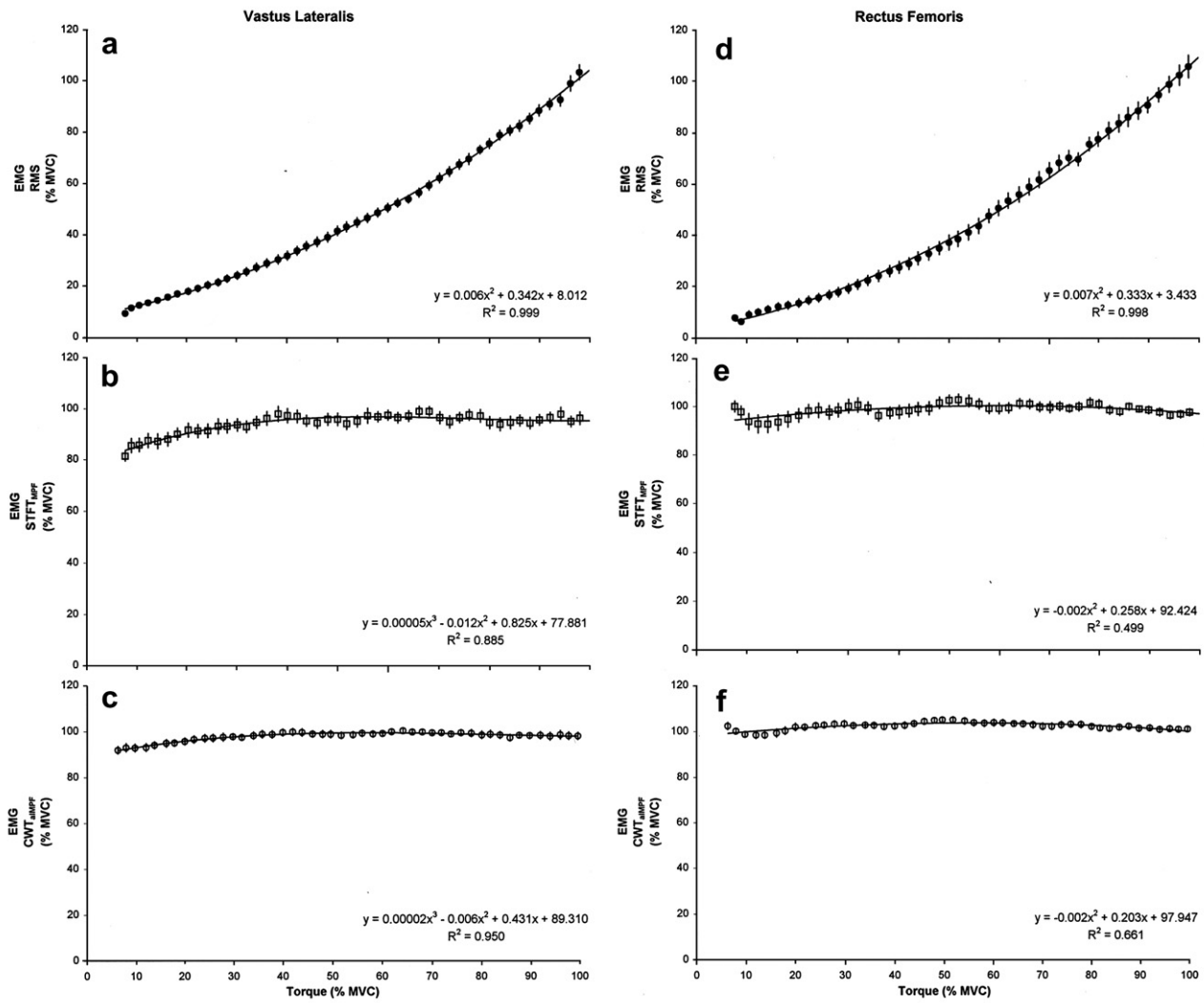


Fig. 4. (a) The relationship between EMG amplitude (RMS, % MVC) and torque (% MVC) for the vastus lateralis (VL) was fit with a quadratic model. (b) The relationship between EMG $STFT_{MPF}$ (% MVC) and torque (% MVC) for the VL was fit with a cubic model. (c) The relationship between EMG CWT_{aIMPF} (% MVC) and torque (% MVC) for the VL was fit with a cubic model. (d) The relationship between EMG amplitude (RMS, % MVC) and torque (% MVC) for the rectus femoris (RF) was fit with a quadratic model. (e) The relationship between EMG $STFT_{MPF}$ (% MVC) and torque (% MVC) for the RF was fit with a quadratic model. (f) The relationship between EMG CWT_{aIMPF} (% MVC) and torque (% MVC) for the RF was fit with a quadratic model. There were no obvious deflection points for either EMG amplitude or frequency across the torque spectrum. Error bars reflect standard errors (SE).

among these responses may be related to factors that influence the MMG amplitude vs. isometric torque relationship during step contractions. These factors include the time windows chosen for signal analysis, the possibility of fatigue during sustained isometric contractions, muscle fiber type characteristics, the joint angle at which the muscle action is performed, and gender-related differences in motor control strategies (Akataki et al., 2003; Beck et al., 2005c; Ebersole et al., 1998, 1999; Maton et al., 1990; Orizio, 1993). Overall, however the MMG amplitude patterns of response have been examined during isometric step contractions for the BB, superficial quadriceps femoris muscles, and two hand muscles, but these patterns have only been examined during isometric ramp contractions for the BB and FDI muscles. Therefore, the findings of the present study extended those of previous studies by characterizing the MMG patterns of response during isometric ramp contractions for the VL and RF muscles.

In the present study, MMG amplitude of the VL did not change from 5% to 28% MVC, increased rapidly from 28% to 78% MVC, and plateaued from 78% to 100% MVC (Fig. 3a). For the RF, MMG amplitude did not change from 5% to 24% MVC, increased rapidly from 24% to 76% MVC, and decreased from 76% to 100% MVC (Fig. 3d). Though the general patterns of response were similar to those recorded from the BB during ramp contractions in previous studies (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003), the second deflection points for the VL and RF in the present study (78% and 76% MVC, respectively) occurred later than for the BB (60–65% MVC) (Akataki et al.,

2004, 2002, 2001, 2003; Orizio et al., 2003). This difference may be due to muscle architecture (fusiform for the BB vs. pennate for the VL and RF) and/or the earlier maximization of motor unit recruitment for the BB (60–65% MVC) compared to the VL or RF (76–78% MVC), which is consistent with the hypothesis that larger muscles rely more on motor unit recruitment than rate coding to increase force production than smaller muscles (Basmajian and DeLuca, 1985; Lawrence and De Luca, 1983). In addition, though the patterns of response for the VL and RF muscles were similar between 5% and 76–78% MVC, the rate of increase in MMG amplitude was greater for the RF than the VL, and there was a decrease from 76% to 100% MVC for the RF, whereas the VL plateaued from 78% to 100%. These differences between the VL and RF may be due to the single joint function of the VL vs. the biarticular aspect of the RF in the seated position (Housh et al., 1996). It is also possible that these differences may be attributable to the thick iliotibial band that covers much of the VL, which may act as a low-pass filter that attenuates the recorded MMG signal at the higher torque levels. The MMG signal for the RF may be less attenuated, since the iliotibial band would be less restrictive over the RF, and the RF may be more compliant in the seated position due to its origin on the anterior, superior iliac spine. In addition, it is also possible the patterns of response for MMG amplitude can be influenced by inter-individual variability in muscle strength and/or muscle fiber type composition (Beck et al., 2005c).

MMG amplitude responses across the force spectrum may reflect the interactions between motor unit recruitment and firing rates that control muscle force production (Akataki et al., 2004; Barry and Cole, 1990; Beck et al., 2005c; Coburn et al., 2005; Orizio, 1993; Orizio et al., 2003). Akataki et al. (2004) suggested that the initial force region below 25% MVC is characterized by slow-twitch motor unit recruitment and rapid firing rates. Therefore, the stable MMG amplitudes during low levels of torque for the VL and RF may be related to the mechanical fusion of the slow-twitch motor units (Akataki et al., 2004, 2002, 2001, 2003), whereas the rapid increases in MMG amplitude (28–78% and 24–76%, respectively) are related to the subsequent recruitment of progressively higher threshold, fast-twitch motor units (Akataki et al., 2004, 2002, 2001, 2003). The plateau and/or decrease near 80% MVC was similar to previous findings (Beck et al., 2005c, 2004; Coburn et al., 2005, 2004; Ebersole et al., 1998, 1999; Matheson et al., 1997; Orizio et al., 1989) and may be related to (a) muscle stiffness, (b) intramuscular fluid pressure, and/or (c) fusion of motor unit twitches at high firing rates (Akataki et al., 2001; Beck et al., 2004; Coburn et al., 2004; Ebersole et al., 1998, 1999; Orizio, 1993; Orizio et al., 1989).

Muscle stiffness is directly related to the number of attached cross-bridges (Ettema and Huijing, 1994) and has been suggested by previous studies to increase with force production (Barry and Cole, 1990; Ford et al., 1981), which may attenuate MMG amplitude at high levels of isometric force production. In theory, a stiffer (less compliant) muscle would not be free to oscillate, which in turn, would result in decreased MMG amplitude. To illustrate, Barry and Cole (1990) examined isometric twitch responses in isolated frog gastrocnemius muscles and observed that muscle stiffness dictated the vibratory frequency of the muscle. As a result, the authors (Barry and Cole, 1990) suggested that MMG could be used to track changes in muscle stiffness during isometric contractions. Therefore, high levels of isometric torque production (which is likely a consequence of twitch fusion) may increase muscle stiffness, and in turn, impair the lateral oscillations of the active muscle fibers, resulting in a plateau and/or decrease in MMG amplitude.

Intramuscular fluid pressure has also been suggested to be a factor that influences the MMG signal (Orizio et al., 1989). Like stiffness, it has been demonstrated (Sejersted et al., 1984) that intramuscular fluid pressure increases with isometric force production. Theoretically, increases in intramuscular pressure would reduce muscle compliance, and in turn, limit the MMG amplitude response (Orizio et al., 1989). However, a recent study by Sogaard et al. (2006) suggested that intramuscular fluid pressure does not attenuate the MMG signal during either ramp or step contractions at near maximal efforts (Sogaard et al., 2006).

Perhaps the primary hypothesis that has been proposed to explain the attenuation of MMG amplitude responses at high force levels is the increase in motor unit firing rate and the eventual fusion of motor unit twitches (Esposito et al., 1996). As all available motor units are recruited (near 60–80% MVC), increases in firing rates are responsible for further increases in torque production. Since surface MMG is the summation of the mechanical activities of individual motor units (Orizio et al., 1996), where the amplitude reflects the number of unfused, activated motor units and their firing rates, it is possible that the plateaus and/or decreases in MMG amplitude above 80% may reflect higher motor unit firing rates during isometric ramp contractions. The higher motor unit firing rates may result in a progressive fusion of motor unit twitches that would increase muscle stiffness and impair active muscle fiber oscillations (Akataki et al., 2004; Akataki et al., 2001; Akataki et al., 2003; Orizio, 1993; Orizio et al., 2003; Sogaard et al., 2006).

4.2. MMG frequency

MMG frequency has been examined in the BB demonstrating rapid increases in MMG frequency to 25–37% MVC, slow increases to 50–56% MVC, and rapid increases to the end of the force spectrum (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003). The FDI, however, responded with a linear increase in MMG frequency to the end of the force spectrum (Akataki et al., 2003). Similar to MMG amplitude, these findings indicated muscle-specific differences between the BB and FDI for the MMG frequency patterns across the force spectrum, which may warrant the examination of this pattern for other muscles. Thus, to our knowledge, the present study was the first to examine the MMG frequency vs. isometric tor-

que relationship during ramp contractions for the VL and RF muscles.

MMG frequency responses have also been examined during isometric step contractions for the BB (Beck et al., 2005c, 2004; Dalton and Stokes, 1993; Esposito et al., 1996; Maton et al., 1990; Orizio et al., 1990) and the superficial muscles of the quadriceps femoris (Coburn et al., 2005; Coburn et al., 2004; Matheson et al., 1997; Shinohara et al., 1998). Shinohara et al. (1998) described this relationship as "flat to increasing" for the VL and "increasing to flat" (p. 291) for the RF based on a visual inspection of the MMG frequency responses. Dalton and Stokes (1993) reported similar findings for the RF which was characterized by a quadratic increase in MMG frequency, with a plateau occurring above 80–90% MVC. Matheson et al. (1997) dichotomized participants into "high-force" and "low-force" levels, and reported that MMG frequency for the RF increased similarly for both groups to 80% MVC, followed by a decrease for the high-force group and an increase for the low-force group. In addition, in one study, Coburn et al. (2004) reported no change in MMG frequency for the vastus medialis to 40% MVC, followed by a rapid increase to 80% MVC, and a slower increase from 80% to 100% MVC, whereas in other studies they reported linear increases in MMG frequency from 10–20% to 100% MVC (Coburn et al., 2005, 2004). Overall, there appear to be differences between the patterns of response for MMG frequency during isometric ramp vs. step contractions, which may indicate that MMG frequency responses are task-specific during non-fatiguing isometric tasks. Therefore, future studies should compare the MMG patterns of response during isometric step vs. ramp contractions.

The results of the present study demonstrated increases in MMG frequency from 5% to 24–28% MVC, a plateau from 24–28% to 76–78% MVC, and further increases from 76–78% to 100% MVC for both the VL and RF muscles (Fig. 3b and e). The frequency of the MMG signal has been proposed to qualitatively reflect the global motor unit firing rate of a muscle (Akataki et al., 2004, 2002, 2001, 2003; Beck et al., 2006b; Coburn et al., 2005, 2004,; Orizio et al., 2003). The initial increases in MMG frequency at low levels of force production have been attributed to newly recruited, slow-twitch motor units and their increasing firing rates Akataki et al., 2004, 2002, 2001, 2003; Erim et al., 1996). The relatively stable MMG frequency responses for the VL and RF in the middle of the force spectrum have been characterized by the recruitment of higher threshold, fast-twitch motor units (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003), which can be seen during the rapid increases in MMG amplitude from 24–28% to 76–78% MVC (Fig. 3a and d). The increases in MMG frequency beyond 76–78% MVC for the VL and RF may have indicated that all motor units were being actively recruited, and thus, their firing rates increased to increase force production (Akataki et al., 2004, 2002, 2001, 2003; Beck et al., 2006b; Coburn et al., 2005, 2004,; Orizio et al., 2003). Overall, it is important to note in the present study that although the deflections points of the cubic relationships for MMG amplitude and frequency were similar (Fig. 3), the patterns surrounding those deflection points were opposite between MMG amplitude and MMG frequency. That is, as MMG amplitude plateaued or decreased (at the beginning and end of the force spectrum), MMG frequency increased. Whereas when MMG amplitude increased (in the middle of the force spectrum), MMG frequency plateaued. These findings, in conjunction with those reported by Akataki et al. (2004) and Orizio et al. (2003), suggested that MMG amplitude and frequency together can provide unique information regarding motor unit recruitment and motor unit firing rates, respectively, during isometric ramp contractions.

4.3. EMG amplitude

The results of the present study showed quadratic, curvilinear increases in EMG amplitude across the force spectrum (Fig. 4a and d), which were consistent with previous isometric ramp and/or step studies (Akasaka et al., 1997, 2004; Coburn et al., 2005; Maton et al., 1990; Woods and Bigland-Ritchie, 1983). Typically, isometric ramp and step contractions are characterized by linear or curvilinear increases in EMG amplitude over the force spectrum (Akasaka et al., 1997, 2004; Bilodeau et al., 2003; Coburn et al., 2005; Ebersole et al., 1999; Esposito et al., 1996; Lariviere et al., 2001; Maton et al., 1990; Woods and Bigland-Ritchie, 1983; Zwarts and Keidel, 1991). EMG amplitude responses from mixed fiber-type muscles have been shown to exhibit curvilinear increases, whereas more uniform fiber-type muscles tend to exhibit linear responses (Woods and Bigland-Ritchie, 1983). In addition, curvilinear increases in EMG amplitude may be due to progressive increases in the firing rates of motor units with fused twitches, causing an increase in EMG amplitude without a related increase in force production (Woods and Bigland-Ritchie, 1983). Due to the fact EMG amplitude is influenced by both motor unit recruitment and firing rate (Basmajian and DeLuca, 1985), it is impossible to distinguish between these two motor control strategies with traditional surface EMG. Therefore, the EMG amplitude vs. isometric torque relationship offers little insight regarding the motor control strategies used to modulate force production other than increases in muscle activation.

4.4. EMG frequency

The results of this study revealed cubic (VL) or quadratic (RF) patterns of response for EMG frequency (Fig. 4b–c and e–f), however, the changes in EMG frequency across the force spectrum were negligible overall. These findings were consistent with previous studies that have reported curvilinear increases in EMG frequency (Akataki et al., 2004; Bilodeau et al., 1991, 2003; Esposito et al., 1996; Lariviere et al., 2001; Maton et al., 1990), although based on the lack of visible changes in Fig. 4b–c and e–f, these results were also

partly consistent with studies that have reported no changes (Akasaka et al., 1997; Coburn et al., 2005) with increasing force levels. A recent study by Bilodeau et al. (2003) reported increases in EMG frequency with force, but, like the present study, the increases were modest. Karlsson and Gerdle (2001) also reported linear increases in EMG center frequency values with increasing force, however, the average R^2 values ranged from 0.25–0.37, thus, again, the changes were marginal. Traditionally, the frequency of the surface EMG signal during isometric muscle actions has been suggested to reflect the average action potential conduction velocities of the activated motor units (Basmajian and DeLuca, 1985; Karlsson and Gerdle, 2001). In addition, Hermens et al. (1992) have hypothesized that the three main physiological parameters that influence EMG frequency are (a) the shapes of the action potentials that are summed to form the EMG signal, (b) the synchronization of action potentials discharged by the active motor units, and, to a lesser extent, (c) the motor unit discharge rate. Therefore, EMG center frequency values are often used to monitor muscle fatigue, due to the fatigue-induced increase in the width of most action potentials and concomitant decrease in action potential conduction velocity (Bigland-Ritchie et al., 1981; Fuglevand et al., 1993). The utility of EMG frequency during non-fatiguing tasks, however, is less clear. It is possible that EMG frequency may be a more appropriate tool to examine muscle fatigue, rather than changes in motor control strategies during non-fatiguing isometric ramp contractions such as those in the present study.

4.5. Wavelet- vs. fourier-based comparisons

Traditionally, Fourier-based algorithms (i.e., fast Fourier or discrete Fourier transforms) have been used to quantify the center frequency values of surface EMG and/or MMG signals using their power density spectra. This can be done using a single epoch or data segment that is chosen to represent the signal, or the signal can be broken into several consecutive subportions where the center frequency of each epoch can be calculated and tracked over time, which is called the STFT. An underlying assumption for all Fourier procedures is that each signal epoch being analyzed remains stationary in the time and frequency domains. The stationarity assumption is violated if there are changes in signal amplitude or frequency within the epoch being analyzed with the Fourier approach. To avoid the underlying assumption of signal stationarity, previous studies have applied alternative algorithms to construct time–frequency representations of surface EMG (Bonato et al., 1996; Karlsson et al., 2001; Karlsson et al., 2000) and MMG (Beck et al., 2006a, 2005a,b) signals, such as the Wigner–Ville transform, the Choi–Williams transform, and most recently, the continuous wavelet transform (CWT). The CWT procedure uses a pre-determined number of dilated and compressed versions of the mother wavelet to detect the low and high frequency components of the signal, respectively (Rioul and Vetterli, 1991). In at least two previous studies, Karlsson and Gerdle (2001, 1999) reported that the CWT, had better accuracy and precision than the STFT for characterizing the EMG frequency patterns of response. However, Bilodeau et al. (1997) reported that local stationarity existed in surface EMG signals recorded during isometric ramp contractions and suggested that standard Fourier algorithms could be used for either step or ramp contractions. To our knowledge, however, no previous studies have compared the CWT and STFT time–frequency representations of EMG or MMG signals during ramp contractions.

Previous studies have compared the frequency responses from EMG and/or MMG signals generated by wavelet- and Fourier-based algorithms using computer synthesized EMG signals (Karlsson et al., 2000), fatiguing contractions of the BB muscle (Beck et al., 2005b; Hostens et al., 2004), and isokinetic muscle actions of the BB (Beck et al., 2006a, 2005a). Despite the conclusions of Karlsson and Gerdle (2001, 1999) that the CWT was better than the STFT, Beck et al. (2006a, 2005a,b) have consistently demonstrated very similar patterns of response for frequency estimates derived from wavelet- vs. Fourier-based procedures. In addition, Bruns (2004) examined electroencephalographic (EEG) signals with the STFT and wavelet transform and reported no differences in spectral content. The results of the present study are consistent with those of Beck et al. (2006a, 2005a,b) and Bruns (2004) and indicated similar time–frequency representations generated by the CWT and STFT for the MMG (Fig. 3b–c and e–f) and EMG (Fig. 4b–c and e–f) signals during isometric ramp contractions. In fact, the CWT_{aIMPF} and $STFT_{MPF}$ patterns were cubic for the MMG signals of the VL and RF and for the EMG signals of the VL, whereas the patterns were both quadratic for the EMG signals of the RF (Table 1). In each case, however, the $STFT_{MPF}$ patterns were more pronounced and exhibited greater variability than the CWT_{aIMPF} patterns, which was evident by the higher R^2 values for the CWT_{aIMPF} (0.661–0.955) than the $STFT_{MPF}$ (0.499–0.945). Though the superiority of either the CWT or STFT could not be determined by this investigation, the deflection points for the MMG patterns were more visible in the $STFT_{MPF}$ graphs. Furthermore, the similarities between the CWT and STFT time–frequency representations suggested that either procedure could be used for EMG or MMG signals recorded during isometric ramp contractions.

4.6. Conclusion

Overall, in conjunction with previous isometric ramp studies (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003), the present study revealed similarities among the MMG and EMG amplitude and frequency responses for the VL and RF muscles when compared to the BB muscle. One exception may be the plateau and/or decrease in MMG amplitude that occurred earlier in the force spectrum for the BB muscle in previous studies (60–65% MVC) (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003) than the

VL and RF muscles in the present study (76–78% MVC). Increases in torque from 5% to 24–28% MVC may be attributed to the recruitment of slow-twitch motor units and their increased firing rates (Akataki et al., 2004, 2002, 2001, 2003; Erim et al., 1996; Orizio et al., 2003), and in the present study, this force region was accompanied by a decrease or plateau in MMG amplitude and an increase in MMG frequency. Increases in torque from 24–28% to 76–78% MVC may be due to the progressive recruitment of higher threshold, fast-twitch motor units, and were accompanied by rapid increases in MMG amplitude and relatively stable MMG frequencies (Akataki et al., 2004, 2002, 2001, 2003; Orizio et al., 2003). Increases in torque from 76–78% to 100% MVC were associated with rapid increases in MMG frequency and may be due to increases in the firing rates of the already active motor units (i.e. rate coding). Therefore, these findings indirectly suggested that MMG amplitude reflected changes in motor unit recruitment, whereas MMG frequency globally reflected changes in motor unit firing rates. At the same time, however, EMG amplitude increased quadratically, and EMG frequency remained relatively stable across the entire force spectrum, which suggested that MMG may offer more unique information regarding motor control strategies that modulate force production during non-fatiguing isometric ramp contractions than traditional surface EMG. Though previous studies have compared EMG responses between isometric ramp vs. step contractions (Akasaka et al., 1997; Bilodeau et al., 1991; Lariviere et al., 2001), it is unclear whether similar patterns could be expected for MMG amplitude and frequency during these two testing modalities. Therefore, future studies are needed to compare the MMG responses during isometric ramp vs. step contractions across the force spectrum. In addition, the present study indicated similar time–frequency representations generated by CWT and STFT procedures. Even though the $STFT_{MPF}$ patterns showed more pronounced changes across the force spectrum than the CWT_{aIMPF} patterns, our findings suggested that either algorithm could be used to track changes in MMG or EMG signal frequency with force.

References

Akasaka K, Onishi H, Momose K, Ihashi K, Yagi R, Handa Y, Hoshimiya N. EMG power spectrum and integrated EMG of ankle plantarflexors during stepwise and ramp contractions. Tohoku J Exp Med 1997;182(3):207–16.

Akataki K, Mita K, Watakabe M. Electromyographic and mechanomyographic estimation of motor unit activation strategy in voluntary force production. Electromyogr Clin Neurophysiol 2004;44(8):489–96.

Akataki K, Mita K, Watakabe M, Ito K. Age-related change in motor unit activation strategy in force production: a mechanomyographic investigation. Muscle Nerve 2002;25(4):505–12.

Akataki K, Mita K, Watakabe M, Itoh K. Mechanomyogram and force relationship during voluntary isometric ramp contractions of the biceps brachii muscle. Eur J Appl Physiol 2001;84(1-2):19–25.

Akataki K, Mita K, Watakabe M, Itoh K. Mechanomyographic responses during voluntary ramp contractions of the human first dorsal interosseous muscle. Eur J Appl Physiol 2003;89(6):520–5.

Barry DT, Cole NM. Muscle sounds are emitted at the resonant frequencies of skeletal muscle. IEEE Trans Biomed Eng 1990;37(5): 525–31.

Basmajian JV, DeLuca CJ. Muscles alive: their functions revealed by electromyography. fifth ed. Baltimore: Williams and Wilkins; 1985.

Beck TW, Housh TJ, Johnson GO, Cramer JT, Weir JP, Coburn JW, et al. Comparison of the fast Fourier transform and continuous wavelet transform for examining mechanomyographic frequency versus eccentric torque relationships. J Neurosci Methods 2006a;150(1):59–66.

Beck TW, Housh TJ, Johnson GO, Cramer JT, Weir JP, Coburn JW, Malek MH. Does the frequency content of the surface mechanomyographic signal reflect motor unit firing rates? A brief review. J Electromyogr Kinesiol 2006.

Beck TW, Housh TJ, Johnson GO, Weir JP, Cramer JT, Coburn JW, Malek MH. Comparison of Fourier and wavelet transform procedures for examining mechanomyographic and electromyographic frequency versus isokinetic torque relationships. Electromyogr Clin Neurophysiol 2005a;45(2):93–103.

Beck TW, Housh TJ, Johnson GO, Weir JP, Cramer JT, Coburn JW, et al. Comparison of Fourier and wavelet transform procedures for examining the mechanomyographic and electromyographic frequency domain responses during fatiguing isokinetic muscle actions of the biceps brachii. J Electromyogr Kinesiol 2005b;15(2):190–9.

Beck TW, Housh TJ, Johnson GO, Weir JP, Cramer JT, Coburn JW, et al. Gender comparisons of mechanomyographic amplitude and mean power frequency versus isometric torque relationships. J Appl Biomech 2005c;21(1):96–109.

Beck TW, Housh TJ, Johnson GO, Weir JP, Cramer JT, Coburn JW, et al. Mechanomyographic amplitude and mean power frequency versus torque relationships during isokinetic and isometric muscle actions of the biceps brachii. J Electromyogr Kinesiol 2004;14(5):555–64.

Beck TW, Housh TJ, Johnson GO, Weir JP, Cramer JT, Coburn JW, et al. Mechanomyographic and electromyographic time and frequency domain responses during submaximal to maximal isokinetic muscle actions of the biceps brachii. Eur J Appl Physiol 2004;92(3):352–9.

Bigland-Ritchie B, Donovan EF, Roussos CS. Conduction velocity and EMG power spectrum changes in fatigue of sustained maximal efforts. J Appl Physiol 1981;51(5):1300–5.

Bilodeau M, Arsenault AB, Gravel D, Bourbonnais D. EMG power spectra of elbow extensors during ramp and step isometric contractions. Eur J Appl Physiol Occup Physiol 1991;63(1):24–8.

Bilodeau M, Cincera M, Arsenault AB, Gravel D. Normality and stationarity of EMG signals of elbow flexor muscles during ramp and step isometric contractions. J Electromyogr Kinesiol 1997;7(2):87–96.

Bilodeau M, Schindler-Ivens S, Williams DM, Chandran R, Sharma SS. EMG frequency content changes with increasing force and during fatigue in the quadriceps femoris muscle of men and women. J Electromyogr Kinesiol 2003;13(1):83–92.

Bonato P, Gagliati G, Knaflitz M. Analysis of myoelectric signals recorded during dynamic contractions. IEEE Med Biol Mag 1996;15:102–11.

Bruns A. Fourier-, Hilbert- and wavelet-based signal analysis: are they really different approaches. J Neurosci Methods 2004;137(2):321–32.

Coburn JW, Housh TJ, Cramer JT, Weir JP, Miller JM, Beck TW, et al. Mechanomyographic and electromyographic responses of the vastus medialis muscle during isometric and concentric muscle actions. J Strength Cond Res 2005;19(2):412–20.

Coburn JW, Housh TJ, Cramer JT, Weir JP, Miller JM, Beck TW, et al. Mechanomyographic time and frequency domain responses of the vastus medialis muscle during submaximal to maximal isometric and isokinetic muscle actions. Electromyogr Clin Neurophysiol 2004;44(4):247–55.

Coburn JW, Housh TJ, Weir JP, Malek MH, Cramer JT, Beck TW, et al. Mechanomyographic responses of the vastus medialis to isometric and eccentric muscle actions. Med Sci Sports Exerc 2004;36(11):1916–22.

Cohen L. Time–frequency analysis. New Jersey: Prentice Hall; 1995.

Dalton PA, Stokes MJ. Frequency of acoustic myography during isometric contraction of fresh and fatigued muscle and during dynamic contractions. Muscle Nerve 1993;16(3):255–61.

Ebersole KT, Housh TJ, Johnson GO, Evetovich TK, Smith DB, Perry SR. The effect of leg flexion angle on the mechanomyographic responses to isometric muscle actions. Eur J Appl Physiol Occup Physiol 1998;8(3):264–9.

Ebersole KT, Housh TJ, Johnson GO, Evetovich TK, Smith DB, Perry SR. MMG and EMG responses of the superficial quadriceps femoris muscles. J Electromyogr Kinesiol 1999;9(3):219–27.

Erim Z, De Luca CJ, Mineo K, Aoki T. Rank-ordered regulation of motor units. Muscle Nerve 1996;19(5):563–73.

Esposito F, Malgrati D, Veicsteinas A, Orizio C. Time and frequency domain analysis of electromyogram and sound myogram in the elderly. Eur J Appl Physiol Occup Physiol 1996;73(6):503–10.

Ettema GJ, Huijing PA. Skeletal muscle stiffness in static and dynamic contractions. J Biomech 1994;27(11):1361–8.

Ford LE, Huxley AF, Simmons RM. The relation between stiffness and filament overlap in stimulated frog muscle fibres. J Physiol 1981;311: 219–49.

Fuglevand AJ, Zackowski KM, Huey KA, Enoka RM. Impairment of neuromuscular propagation during human fatiguing contractions at submaximal forces. J Physiol 1993;460:549–72.

Hermens HJ, Van Bruggen TAM, Baten CTM, Rutten WLC, Boom HBK. The median frequency of the surface EMG power spectrum in relation to motor unit firing and action potential properties. J Electromyogr Kinesiol 1992;2:15–25.

Hostens I, Seghers J, Spaepen A, Ramon H. Validation of the wavelet spectral estimation technique in biceps brachii and brachioradialis fatigue assessment during prolonged low-level static and dynamic contractions. J Electromyogr Kinesiol 2004;14(2):205–15.

Housh TJ, deVries HA, Johnson GO, Evans SA, Housh DJ, Stout JR, et al. Neuromuscular fatigue thresholds of the vastus lateralis, vastus medialis and rectus femoris muscles. Electromyogr Clin Neurophysiol 1996;36(4):247–55.

Karlsson JS, Gerdle B, Akay M. Analyzing surface myoelectric signals recorded during isokinetic contractions. IEEE Eng Med Biol Mag 2001;20(6):97–105.

Karlsson S, Gerdle B. Mean frequency and signal amplitude of the surface EMG of the quadriceps muscles increase with increasing torque – a study using the continuous wavelet transform. J Electromyogr Kinesiol 2001;11(2):131–40.

Karlsson S, Yu J, Akay M. Enhancement of spectral analysis of myoelectric signals during static contractions using wavelet methods. IEEE Trans Biomed Eng 1999;46(6):670–84.

Karlsson S, Yu J, Akay M. Time–frequency analysis of myoelectric signals during dynamic contractions: a comparative study. IEEE Trans Biomed Eng 2000;47(2):228–38.

Kossev A, Christova P. Discharge pattern of human motor units during dynamic concentric and eccentric contractions. Electroencephalogr Clin Neurophysiol 1998;109(3):245–55.

Lammert O, Jorgensen F, Kwatny HG. Accelerometermyography (AMG) I: method for measuring mechanical vibrations from isometrically contracted muscles. Baltimore: University Park; 1976.

Lariviere C, Arsenault AB, Gravel D, Gagnon D, Loisel P. Effect of step and ramp static contractions on the median frequency of electromyograms of back muscles in humans. Eur J Appl Physiol 2001;85(6):552–9.

Lawrence JH, De Luca CJ. Myoelectric signal versus force relationship in different human muscles. J Appl Physiol 1983;54(6):1653–9.

Matheson GO, Maffey-Ward L, Mooney M, Ladly K, Fung T, Zhang YT. Vibromyography as a quantitative measure of muscle force production. Scand J Rehabil Med 1997;29(1):29–35.

Maton B, Petitjean M, Cnockaert JC. Phonomyogram and electromyogram relationships with isometric force reinvestigated in man. Eur J Appl Physiol Occup Physiol 1990;60(3):194–201.

Orizio C. Muscle sound: bases for the introduction of a mechanomyographic signal in muscle studies. Crit Rev Biomed Eng 1993;21(3):201–43.

Orizio C, Gobbo M, Diemont B, Esposito F, Veicsteinas A. The surface mechanomyogram as a tool to describe the influence of fatigue on biceps brachii motor unit activation strategy. Historical basis and novel evidence. Eur J Appl Physiol 2003;90(3-4):326–36.

Orizio C, Liberati D, Locatelli C, De Grandis D, Veicsteinas A. Surface mechanomyogram reflects muscle fibres twitches summation. J Biomech 1996;29(4):475–81.

Orizio C, Perini R, Diemont B, Maranzana Figini M, Veicsteinas A. Spectral analysis of muscular sound during isometric contraction of biceps brachii. J Appl Physiol 1990;68(2):508–12.

Orizio C, Perini R, Veicsteinas A. Changes of muscular sound during sustained isometric contraction up to exhaustion. J Appl Physiol 1989;66(4):1593–8.

Orizio C, Perini R, Veicsteinas A. Muscular sound and force relationship during isometric contraction in man. Eur J Appl Physiol Occup Physiol 1989;58(5):528–33.

Petitjean M, Maton B, Cnockaert JC. Evaluation of human dynamic contraction by phonomyography. J Appl Physiol 1992;73(6):2567–73.

Rioul O, Vetterli M. Wavelets and signal processing. IEEE Signal Process Mag 1991:14–38.

Sejersted OM, Hargens AR, Kardel KR, Blom P, Jensen O, Hermansen L. Intramuscular fluid pressure during isometric contraction of human skeletal muscle. J Appl Physiol 1984;56(2):287–95.

Shinohara M, Kouzaki M, Yoshihisa T, Fukunaga T. Mechanomyogram from the different heads of the quadriceps muscle during incremental knee extension. Eur J Appl Physiol Occup Physiol 1998;78(4):289–95.

Sogaard K, Orizio C, Sjogaard G. Surface mechanomyogram amplitude is not attenuated by intramuscular pressure. Eur J Appl Physiol 2006;96(2):178–84.

Stokes MJ, Cooper RG. Muscle sounds during voluntary and stimulated contractions of the human adductor pollicis muscle. J Appl Physiol 1992;72(5):1908–13.

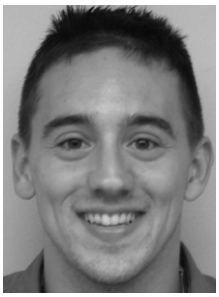
Stokes MJ, Dalton PA. Acoustic myographic activity increases linearly up to maximal voluntary isometric force in the human quadriceps muscle. J Neurol Sci 1991;101(2):163–7.

Stokes MJ, Dalton PA. Acoustic myography for investigating human skeletal muscle fatigue. J Appl Physiol 1991;71(4):1422–6.

Stulen FB, DeLuca CJ. Frequency parameters of the myoelectric signal as a measure of muscle conduction velocity. IEEE Trans Biomed Eng 1981;28(7):515–23.

Woods JJ, Bigland-Ritchie B. Linear and non-linear surface EMG/force relationships in human muscles. An anatomical/functional argument for the existence of both. Am J Phys Med 1983;62(6):287–99.

Zwarts MJ, Keidel M. Relationship between electrical and vibratory output of muscle during voluntary contraction and fatigue. Muscle Nerve 1991;14(8):756–61.



Eric D. Ryan is currently a doctoral student in the exercise physiology department at the University of Oklahoma under the mentorship of Dr. Joel Cramer. He received a BS (2003) degree in exercise science from Tulane University, New Orleans, Louisiana, and a MS (2005) degree from Florida Atlantic University, Davie, Florida in exercise physiology. He is a member of the American College of Sports Medicine (ACSM), American College of Occupational and Environmental Medicine (ACOEM), and National Strength and Conditioning Association (NSCA). His primary research interests include the non-invasive assessment of muscle function and the acute effects of different stretching routines on parameters of muscle strength and flexibility.

Dr. Cramer is an Assistant Professor and Director of the Biophysics Laboratory in the Department of Health and Exercise Science and *Adjunct Assistant Professor* in Industrial Engineering at the University of Oklahoma. Dr. Cramer received a bachelor's degree in Exercise Science from Creighton University in 1997 and a master's and doctorate in Exercise Physiology from the University of Nebraska-Lincoln in 2001 and 2003, respectively, under Dr. Terry J. Housh. Joel is a member of the National Strength and Conditioning Association (NSCA), the American College of Sports Medicine (ACSM), and the International Society of Electrophysiology and Kinesiology (ISEK). He studies the noninvasive assessment of muscle function with surface electromyography (EMG) and mechanomyography (MMG). In 2005, Dr. Cramer received the Young Investigator of the Year Award from the NSCA.

Alison Egan is currently working on her Master's degree in Exercise Physiology at the University of Oklahoma. She received a Bachelor's degree from the University of Wisconsin—La Crosse in December 2004 after completing an internship in Sports Physiology/Biomechanics at the United States Olympic Training Center in Chula Vista, California.

Michael Hartman, MS, is a doctoral candidate in exercise physiology within the Department of Health and Exercise Science at the University of Oklahoma. He holds a bachelor of science in exercise science from Towson University, and a master of science in kinesiology with a concentration in exercise physiology from Midwestern State University. Previously, Michael served as a sport physiology research assistant in the Division of Coaching and Sport Science at the United States Olympic Committee.

Trent J. Herda graduated with a Bachelor of Science Degree in Exercise Science from the University of Sioux Falls in December 2005. He is currently a graduate assistant in the Biophysics Laboratory in the Exercise Physiology Department at the University of Oklahoma. His research interests include the acute effects of various stretching modalities on human performance and neuromuscular function.